



Institute of Technology University of Moratuwa

Community-Based Ocean & Coastal Cleanup Management System

Student Details:

KMNI Ranasinghe -23IT0520

VPC Rasanga-23IT0521

SMR Kaveeshwara-23IT0492

MDB Amarakoon-23IT0452

MRS Nishadini-23IT0512

HGMN Kodiweera-23IT0496

Supervised by:

Mrs. Shashika Ruwanmini

Division of Information Technology
Institute of Technology University of Moratuwa

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Abstract

One of the major environmental concerns in Sri Lanka is marine pollution. Though environmental concern is on the rise among Sri Lankans, there is no platform that enables citizens, volunteers, divers, government agencies, and organizations to collaborate effectively in order to manage ocean and coastal cleanup activities. Therefore, the proposed "Community Based Ocean & Coastal Cleanup Management System" is an effective solution that utilizes technology in addressing environmental concerns in Sri Lanka.

This system allows users to report polluted areas on the coast using photographic and GPS technologies. The system also enables verification of reported pollution sites by calculating the trust percentage. Once verified, users can then obtain approval from local government agencies. Other features include alerts for volunteers and divers at polluted sites, monitoring project progress, employment opportunities for divers with NGOs and tourism agencies, and monitoring pollution levels. The application is developed using multi-tier architecture with Spring Boot and React.js. This project aims to significantly improve environmental protection in Sri Lanka by empowering communities to collaborate effectively in order to achieve environmental impact.

1.Introduction

Marine pollution has been identified as one of the most important environmental concerns for nations around the world. Sri Lanka, with a long coastline of 1,340 km and rich biodiversity, is highly susceptible to the adverse effects of sea and ocean pollution. The coastal areas of Sri Lanka support one-third of the total population and form the basis for important sectors such as tourism, fishing, and shipping [2]. However, rapid urbanization and lack of proper waste management and response mechanisms have worsened the situation.

The Marine Environment Protection Authority (MEPA) states that Sri Lanka produces about 1.6 million tons of plastic waste every year [2]. The recent maritime accident involving the vessel X-Press Pearl in 2021 highlighted the sensitivity of Sri Lanka's marine ecosystem, spilling about 1,680 tons of plastic and harmful chemicals into the Indian Ocean [6]. In addition to major accidents, day-to-day pollution from coastal populations and land-based waste also affects the marine ecosystem.

Despite increased environmental consciousness and the willingness of citizens to contribute toward solutions, there is no integrated digital platform in Sri Lanka to effectively connect all stakeholders for coastal conservation. Citizens aware of pollution have no avenue to report information. Volunteers engaged in cleaning activities have no official platform to coordinate activities or measure success. Professional divers willing to contribute their expertise are not connected with opportunities. Government authorities receive information through informal channels, making it difficult to authenticate information and prioritize responses [3].

The Community-Based Ocean & Coastal Cleanup Management System is proposed as an innovative technological solution to fill these gaps, serving as a central platform for identifying, verifying, and managing coastal cleanup activities.

2. Background and Motivation

Sri Lanka's coastal and marine ecosystems face significant threats from pollution. These regions support one-third of the population and are vital for tourism, fishing, and shipping. The National Environmental Action Plan 2022–2030 identifies marine and coastal pollution as a critical concern, but effective solutions are hindered by poor stakeholder coordination, lack of centralized data, and limited public participation [2]. Furthermore, according to MEPA (2025), approval is required for “any project or activity implemented within Sri Lanka's territorial waters, maritime zones, foreshore, or coastal areas,” highlighting the need for proper coordination and regulatory oversight in managing coastal environments [2].

2.1 Motivating Factors

Several key factors highlight the critical need for a digital coastal cleanup management system:

- **Fragmented Reporting Mechanisms:** Currently, there is no standard system for citizens to report polluted areas. The only means of reporting are through social media, calling authorities, or not reporting at all. This lack of an efficient reporting system discourages public participation in cleanup activities.
- **Lack of Verification Systems:** The absence of proper verification systems makes it difficult for authorities to differentiate between correct and false information. Citizens may also doubt the authenticity of the information they share.
- **Inefficient Coordination of Volunteers:** Cleanup activities are often coordinated through informal means such as WhatsApp groups and Facebook posts, lacking tools to assess and evaluate activities. Volunteers may travel long distances only to find poor coordination.
- **Limited Diver Engagement:** Recreational and professional divers willing to participate in underwater cleanup activities lack a platform to find assignments that suit their certification and geographical needs.
- **Lack of Government Coordination:** Cleanup projects need approvals from government departments, including the Coastal Conservation Department and

MEPA. The manual approval process is time-consuming and lacks transparency, causing delays from days to weeks.

- **No System to Monitor Overall Impact:** Without a single system, it is hard to measure the overall impact of cleanups, monitor pollution trends, and efficiently utilize resources for specific areas.

2.2 Background of the Proposed Solution

To overcome these challenges, the proposed system offers an extensive digital platform with the following functionalities:

- **Community-Driven Reporting:** Any registered user can report polluted areas with images, videos, and GPS coordinates, creating a database accessible to all stakeholders.
- **Collaborative Verification:** Community-driven verification through voting and discussion establishes trust through collective intelligence with threshold-based verification logic.
- **Structured Project Management:** A workflow for cleanup projects from initial reporting to government approvals, volunteer mobilization, and progress tracking.
- **Location-Based Alerting:** Alerts to volunteers and divers near cleanup project locations, with alert radius extensions if initial responses are insufficient.
- **Government Integration:** Digital interaction for relevant government authorities to review and approve projects with comments and instructions.
- **Diver Opportunity Platform:** A portfolio system where divers list their certifications, experience, equipment, and regions of interest to connect with NGOs, tourism operators, and marine institutions.

3. Problem in Brief

Sri Lanka faces increasing pollution problems in its coastal and marine environment. However, the current approach to solving this environmental crisis is based on fragmented and manual processes that are inefficient and often ineffective. In the absence of a centralized electronic system, operational challenges are faced by all stakeholders in coastal conservation.

3.1 The current manual process creates issues for the following:

- **Community Members and Citizens:** Citizens seeing pollution have no efficient way to report the matter. Even if they report through social media or phone calls, there is no feedback mechanism to know if action has been taken, discouraging further participation in environmental conservation.
- **Volunteers and Environmental Enthusiasts:** People willing to contribute through cleaning activities have no mechanism to find opportunities, monitor their activities, or measure the impact of their efforts.
- **Divers:** Professional and recreational divers with skills to assist in underwater cleanup have no platform to access opportunities that fit their skill level and experience, resulting in loss of valuable expertise.
- **Government Authorities:** Local government, MEPA, and other agencies receive information from various sources but lack a single platform for efficient resource allocation to priority areas.
- **NGOs and Environmental Organizations:** Organizations working for coastal conservation face difficulties finding priority areas, mobilizing volunteers, liaising with government authorities, and measuring impact for funding and support.

3.2 What happens if the problem is not solved?

- **Accelerated Environmental Degradation:** Pollution will continue to increase, affecting marine life, biodiversity, and human health through contaminated seafood and water supplies.

- **Wasted Volunteer Efforts:** Volunteers may conduct uncoordinated activities without proper planning, resulting in inefficient resource utilization, potential accidents, and missed critical pollution areas.
- **Missed Opportunities for Professional Engagement:** Opportunities for divers to engage in environmental conservation remain unfilled, and organizations seeking diving skills cannot access required human capital.
- **Ineffective Government Response:** The government cannot see the full extent of pollution or affected areas, leading to inadequate response and policy development focused on easily visible rather than most harmful pollution.
- **Loss of Public Trust:** When people report pollution and see no action taken, public trust in environmental protection institutions erodes, creating a vicious cycle of environmental deterioration and reduced public action.

3.3 Why is it important to fix this problem?

- **Environmental Protection:** Coastal and marine environments provide essential ecosystem services including food security, tourism, coastal protection, and biodiversity conservation. Managing cleanup activities is essential for preserving these services for future generations.
- **Public Health Protection:** Coastal pollution can result in seafood contamination, disease carrier breeding, and toxic chemical release. Timely cleanup activities protect resident health and reduce illness treatment costs.
- **Economic Sustainability:** Tourism and fishing sectors are vital to Sri Lanka's economy, supporting countless livelihoods. Beach cleanliness is essential for continued tourism revenue generation.
- **Community Empowerment:** This system provides a structured approach for the public to become active participants in environmental protection rather than passive observers, building social capital.
- **Efficient Resource Allocation:** The system provides valuable data on pollution patterns and cleaning activities for efficient resource allocation by government and non-government organizations.

4.Aim & Objectives

4.1 Aim

To develop a Community-Based Ocean & Coastal Cleanup Management System that connects citizens, volunteers, divers, government authorities, and organizations for the identification, verification, and management of coastal pollution cleanup activities in Sri Lanka.

4.2 Objectives

- **Gather Requirements:** Consult with all stakeholder groups (community members, volunteers, divers, government officers, organizations) to define system requirements.
- **Build Reporting & Verification:** Implement pollution reporting with photos, videos, GPS, community voting, trust percentage calculation, and 75% threshold-based verification to ensure the accuracy and reliability of reported data.
- **Create Government Approval Workflow:** Develop digital approval module for government authority officers to review, approve/reject cleanup requests, and record comments.
- **Implement Location-Based Alerts:** Develop notification system to detect nearby users, send alerts with map locations, and escalate alerts when no response is received.
- **Build Project Monitoring:** Enable progress tracking with before/during/after images, completion percentages, and cleanup results recording to monitor cleanup effectiveness and ensure transparency of activities.
- **Create Diver Opportunity Module:** Connect divers with NGOs, tourism operators, and marine institutions for employment and job opportunities.
- **Develop Analytics Dashboard:** Provide reporting on polluted sites, verified incidents, volunteer participation, and regional pollution trends to support data-driven decision-making and improve environmental management strategies.

- **Ensure Security & Scalability:** Implement access controls, data validation, and Multi-tier architecture for data integrity and system growth.

5.Project Scope

This system is designed to help identify polluted ocean and coastal locations through community reports, verify those reports, organize volunteer cleanup projects, coordinate nearby divers and community members, monitor project progress, and connect relevant authorities when approval or intervention is needed. It also creates a social platform for environmental protection and can further support employment opportunities for divers.

System Users of the Proposed System

- General Community Users (Registered users with full access and non-registered users with limited access.)
- Volunteer Divers
- Government Authority Officers
- System Administrator
- Organizations (possible sub-categories: NGOs, Tourism, Marine Institutions)

5.1 In-Scope (Features & Functionalities)

Module 1: User Registration and Profile Management Module

This module handles user account creation and profile maintenance. Different user roles can register in the system. Users can manage personal details, contact information, current location, and skills. Divers can additionally include certification level, diving experience, equipment availability, and preferred working regions.

Main functions:

- User registration and login
- Role-based access
- Profile editing (diver skill and certification details)
- Location and contact details
- User availability status.

Module 2: Pollution Reporting and Incident Submission Module

This is a core module of the system. Any registered user can report a polluted ocean or coastal area by uploading photographs, videos, location details, and a short description. This module is the entry point of the cleanup process.

Main functions:

- Upload polluted location photos and videos
- Add description of the issue,
- Capture map coordinates/GPS location
- Submit date and time
- Attach multiple images or evidence.

Module 3: Community Verification and Trust Validation Module

After a pollution report is submitted, the system allows other users to verify whether the report seems genuine. Users can vote (true/false), react, and comment. Based on the percentage of positive confirmations, the system decides whether the report is community-verified. This prevents misuse of the system.

Main functions:

- True/false marking by users
- User comments and discussion
- Confidence/trust percentage calculation
- Threshold-based verification logic (75% threshold)
- Admin review when trust percentage is below threshold
- Fake report detection support.

Module 4: Admin Review and Moderation Module

If a post is not sufficiently verified by the community, the administrator can manually inspect the report. The admin can accept, reject, or request further clarification. The admin also manages users, controls abuse, and maintains system reliability.

Main functions:

- Review unverified or suspicious posts
- Approve/reject reports

- Manage abusive users or false posts
- Monitor community activities
- Handle complaints and disputes
- Maintain overall system quality

Module 5: Government Body Coordination and Approval Module

Marine Environmental Protection Authority (MEPA) officers can register with the system. This module supports the digital approval workflow for cleanup projects requiring government permission.

Main functions:

- Forward verified projects to relevant government body
- Allow authority officers to review requests
- Approve or reject cleanup permission
- Digital approval request submission
- Approval/rejection/pending workflow
- Record authority comments and instructions
- Notify admin about approval status
- Authority dashboard (Future Scope)
- Approval notifications.

Module 6: Location-Based Alert and Notification Module

Once a pollution post is verified, the system sends alerts to users near that location. Nearby divers, volunteers, and community members receive the message first. If nobody accepts within a certain time, the alert is widened to a larger area and finally to users around the country.

Main functions:

- Detect nearby registered users
- Send notifications with map location
- Escalate alerts when no one responds
- Send reminders to potential volunteers
- Notify project progress updates, inform authorities if needed.

Module 7: Project Progress Monitoring Module

This module tracks the completion of each cleanup project. The project owner can upload before, during, and after photos, progress descriptions, task completion percentages, and cleanup outcomes. This creates transparency and accountability.

Main functions:

- Upload periodic project updates
- Add progress images and notes
- Update status timeline
- Measure completion percentage
- Display before/after comparison
- Record cleanup results

Module 8: Diver Employment and Opportunity Module

Divers involved in community service may also benefit from paid opportunities later. This module connects divers with organizations needing diving services.

Main functions:

- Diver profile and skill showcase
- Post job opportunities for divers
- Participated charity projects with marks given by project owner
- Connect divers with NGOs/tourism/marine organizations/research teams
- Notification system about job opportunities
- Record past volunteer experience as portfolio

Module 9: Reporting and Analytics Module

This module helps administrators and authorities monitor overall system performance and environmental impact.

Main functions:

- Number of reported polluted sites
- Number of verified incidents
- Number of active and completed projects
- Volunteer participation statistics
- Region-wise pollution trends

5.2 Out of Scope (Exclusions & Limitations)

- **Physical Cleanup Operations:** Physical cleanup activities will not be executed by this system; they will be coordinated and tracked within the system.
- **Emergency Response:** The system is for planned cleanup projects, not for emergency responses to pollution or hazardous material situations.
- **Integration with International Systems:** The system is designed only for Sri Lankan environments and will not be integrated with international environmental monitoring systems.
- **Native Mobile Application:** The system will be web-based; mobile support will be through mobile web browsers. Native mobile applications for iOS and Android are considered for future development.
- **Real-Time GPS Tracking:** Although the system records location information for report generation and notifications, real-time tracking of volunteers during cleanup activities is not included.
- **Financial Transactions:** The system will not handle any financial transactions between organizations and divers.
- **Legal Enforcement:** The system will not have legal enforcement features or integration with law enforcement agencies.

5.3 Assumptions

- **Data Availability:** Users will provide accurate location and pollution information during reporting. Organizations will provide accurate details for job opportunities.
- **Internet Connectivity:** Users will have internet access to use the web-based system through standard broadband connections.
- **User Training and Adoption:** Government authorities, organizations, and users will adopt the system. Training opportunities will be available.

- **Legal and Compliance Requirements:** The system will comply with Sri Lankan legal and regulatory requirements regarding data protection and environmental concerns.
- **Geographic Coverage:** The system will initially focus on Sri Lankan coastal areas, with possible extension to other areas in subsequent development phases.
- **External API Usage:** The system will use external APIs (e.g., geocoding APIs for location validation, and map APIs for displaying pollution locations) to extract relevant environmental data and enhance the accuracy of pollution reports and cleanup planning.
- **API Availability:** Required external services will be available and accessible along with their terms of service.

6.Methodology, Implementation Considerations

6.1 Software Development Methodology

Our project timeline has been designed as Waterfall stages. However, even though the project timeline is provided as a Waterfall model, in each respective phase the Agile characteristics are expected to be used by planning the tasks in small sprints [1].

Waterfall Stages with Agile Characteristics:

Waterfall Stage	Agile Implementation
Requirements Gathering	Sprints for each stakeholder group (2-3 weeks per sprint)
System Design	Sprints for each module design
Development	2-3 weeks sprints per module
Testing	Continuous testing within each sprint
Deployment	Iterative deployment of completed modules

Table 1 : Waterfall Stages with Agile Characteristics

Benefits of this approach:

- Structured Phases: Clear milestones and deliverables as per Waterfall
- Iterative Development: Feedback from stakeholders during each phase
- Flexibility: Easy incorporation of changes within phases
- Frequent Deliverables: Features delivered in short sprints (2-3 weeks)

- Collaboration: Regular communication among developers and stakeholders
- Risk Management: Early detection through continuous testing and integration

6.2 Deployment Strategy

The system will be deployed as a cloud-based web application:

- Cloud Hosting: AWS, Google Cloud, or Azure for scalability and availability
- Remote Accessibility: Accessible via any internet-enabled device
- Centralized Data Management: Consistent, real-time data access
- Scalability: Ability to add users, data, and future mobile applications
- Backup & Recovery: Business continuity and data integrity mechanisms

6.3 System Architecture

The system will adhere to a multi-Tier Architecture to ensure scalability, reliability, and maintainability.

- **Scalability:** The multiple layers (Presentation, API/Communication, Application, Data, and Integration) can be scaled independently based on system requirements.
- **Reliability:** Separation into multiple tiers helps isolate faults, ensuring that failures in one layer do not affect the entire system.
- **Performance:** Structured interaction between different tiers improves overall system efficiency and response time.
- **Maintainability:** Each tier can be updated or modified independently, simplifying system maintenance and future enhancements
- **Separation of Concerns:** Each tier is responsible for a specific function, improving system organization and reducing complexity.

Justification of Technologies Used in the Proposed System

Technology	Purpose	Justification
Spring Boot	Backend (application tier) development using Spring Boot with microservices architecture	Ensures scalability, security, and rapid development with built-in support for REST APIs [5].
React.js	Frontend (User Interface)	Provides a fast, responsive UI, supports reusable components, and integrates well with REST APIs [4].
MySQL	Database	Relational database with strong ACID compliance, ensuring data integrity for pollution reports, and user information.
Docker Kubernetes	Containerization	Enables efficient deployment & scaling in cloud environments

Table 2 : Technology stack used in Proposed System

7. Timeline

Timeframe	Task
2 weeks	Background and Literature Study
2 weeks	Requirement Gathering and Analysis
2 weeks	Draft SRS Preparation
1 week	SRS Report Submission
2 weeks	System Design (Design Diagrams + Review)
3 weeks	Chapter Writing and Draft Report Preparation
1 week	Final Presentation Preparation
1 week	Final Report Submission

Table 3 : Timeline

8. Work Breakdown

WBS Id	Task Name	Description	Assigned To
1	Project Initiation	Defining project scope, objectives, and requirements.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
1.1	Requirement Gathering	Conduct meetings with stakeholders to collect requirements.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
1.2	Project Proposal	Prepare and submit project proposal for approval.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
2	System Design and System Architecture	Designing the multi-tier architecture and defining microservices-based backend using Spring Boot.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
2.1	UI/UX design	Creating wireframes and UI mock-ups.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera

3	Frontend Development	Developing the user interface using React.js.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
4	Backend Development	Developing the server-side logic with Spring Boot.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
5	Testing & Quality Assurance	Ensuring the system is bug-free and meets requirements.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
6	Deployment & Implementation	Making the system live for use.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera
7	User Training and Documentation	Training users on system usage.	KMNI Ranasinghe VPC Rasanga SMR Kaveeshwara MDB Amarakoon MRS Nishadini HGMN Kodiweera

Table 4 : Work Breakdown

9. References

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